

VALUE BEFORE SILENCE

A dual-model CLTV framework for Olist's next-quarter planning and precision win-back

Sabina Gallo 535785

AI for Communication and Marketing

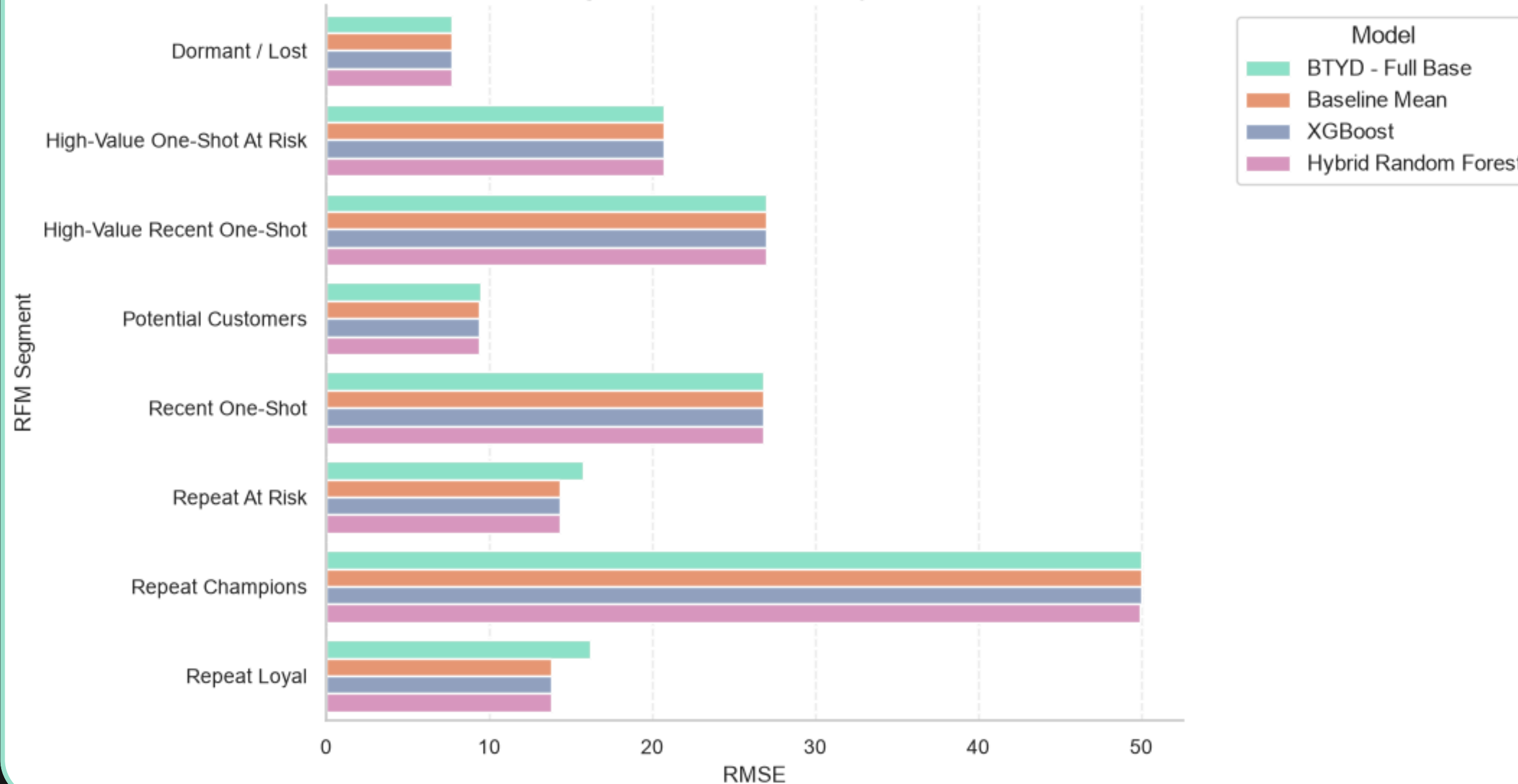
Academic Year 2025/2026



THE WINNER CHANGES WITH THE CUSTOMER

Model performance varies across segments and no single approach dominates every decision

Segment-Level RMSE by Model



20.1178

Lowest RMSE

Hybrid Random Forest

30.72%

Top-decile value capture

Hybrid XGBoost

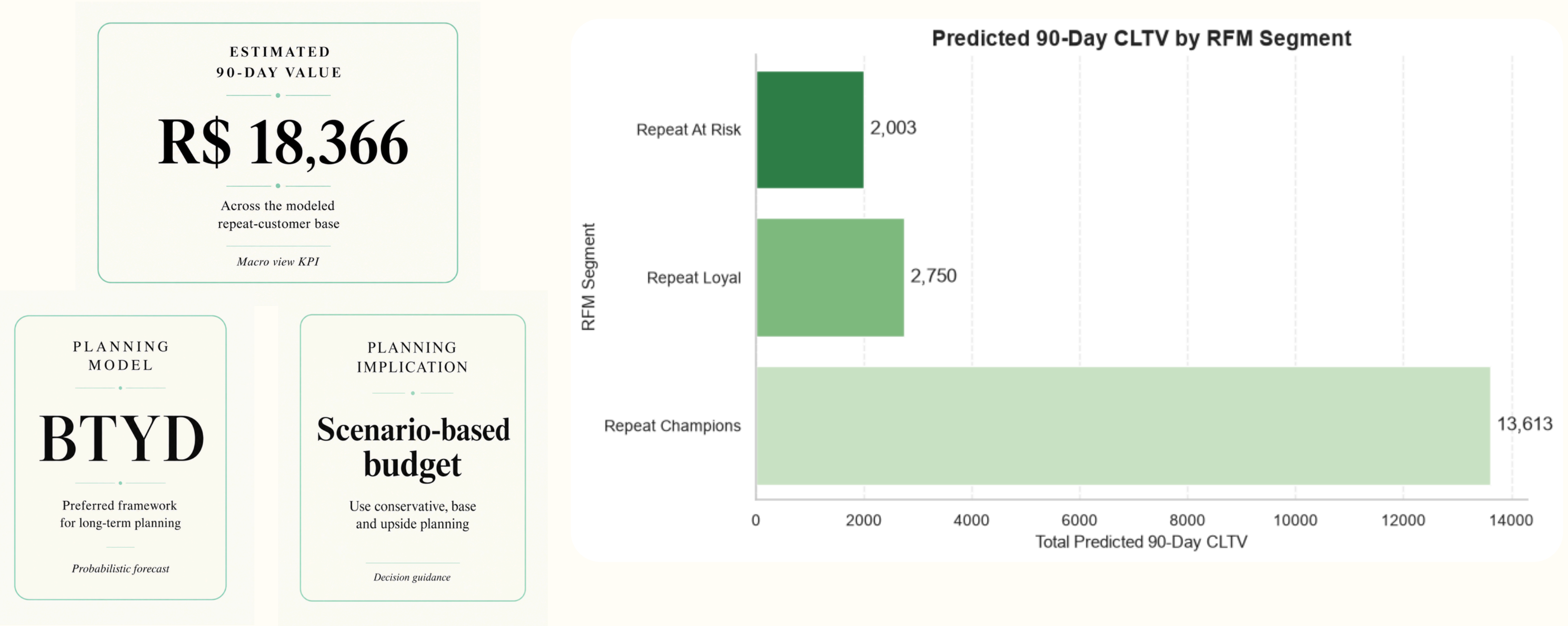
Negative R²

Absolute revenue remains
difficult to predict

Hybrid Random Forest provides the strongest numerical accuracy, while Hybrid XGBoost produces the most effective customer ranking. Model choice therefore depends on the business decision.

A FORECAST IS NOT YET A BUDGET

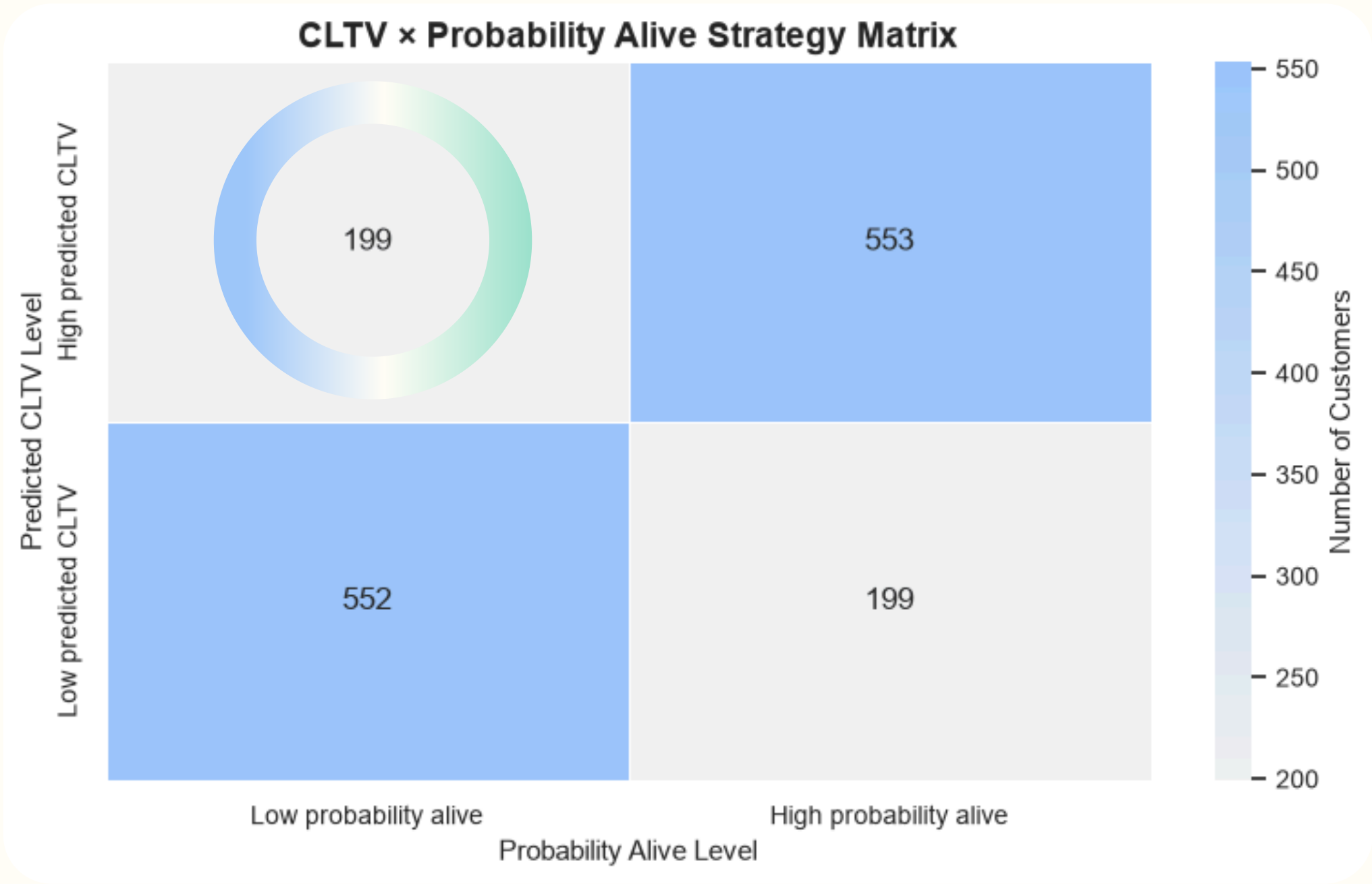
BTYD reveals where future value is concentrated, but current calibration calls for scenario-based planning



BTYD is better for long-term budget forecasting because it is a probabilistic model designed to estimate future purchasing behavior and latent customer inactivity in a non-contractual setting.

FIND THE VALUE ABOUT TO DISAPPEAR

High predicted CLTV becomes actionable when combined with a low probability of remaining active



PRIORITY
SEGMENT

**199
customers**

High predicted CLTV
× Low probability alive

Win-back target

WIN-BACK
CAMPAIGN

**Silent Value
Recovery**

Recover valuable customers
before silence becomes permanent

Micro view strategy

CONTACT
PLAN

**Personalized email
+ CRM retargeting**

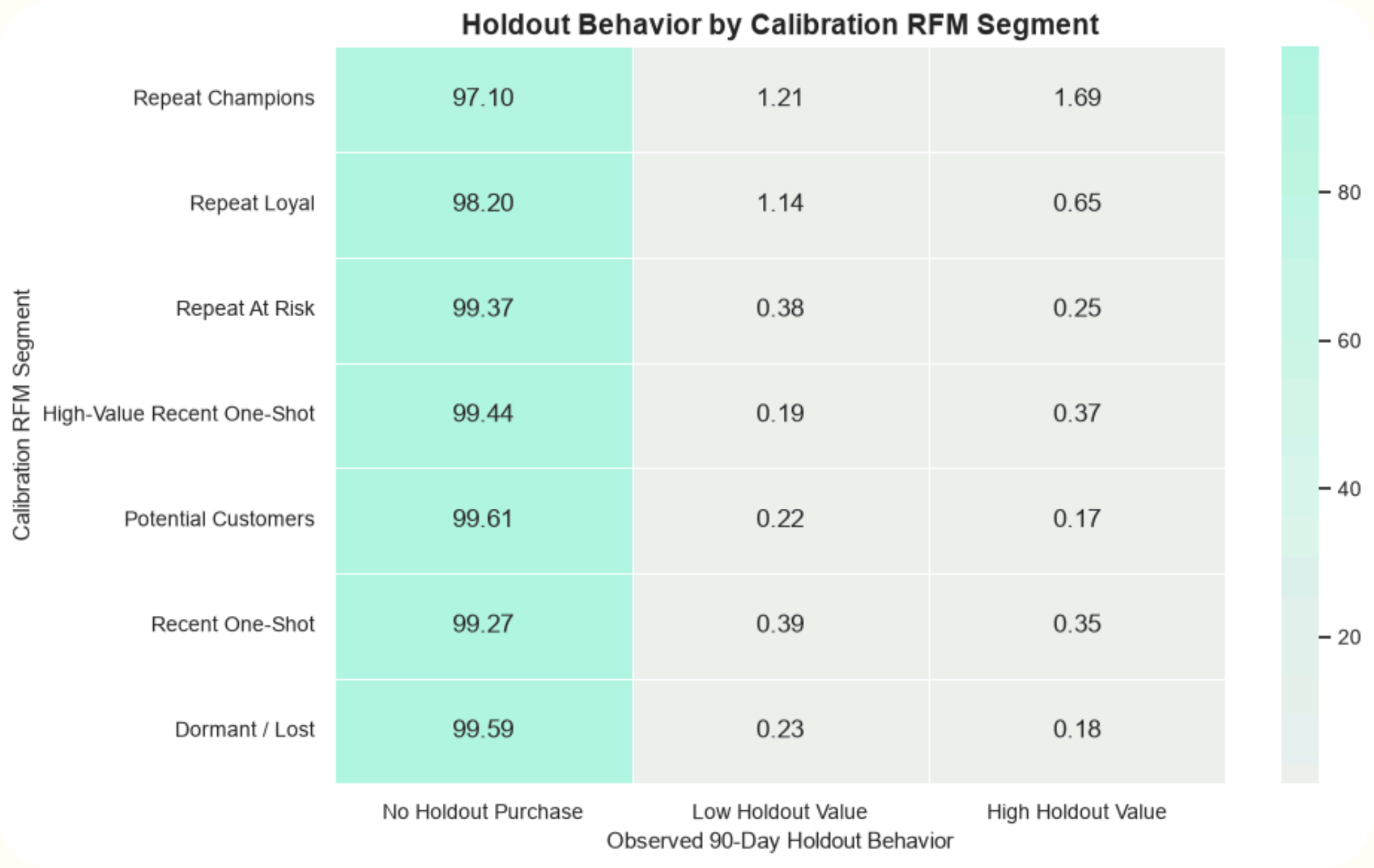
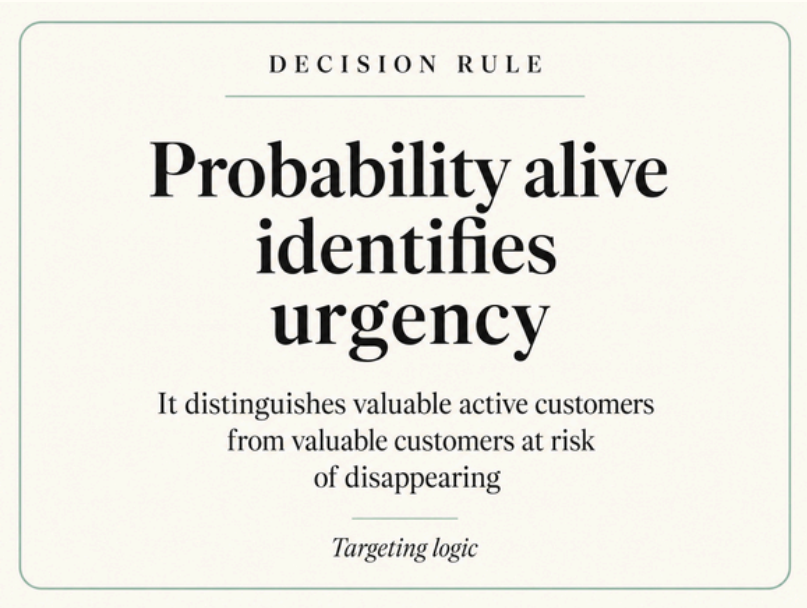
Start with a relevant reminder
and follow non-openers with retargeting

Activation approach

Do not target inactivity alone: prioritize the customers whose disappearance would destroy the greatest future value.

THE SILENT CHAMPION PARADOX

Holdout behavior shows why historical segmentation must be combined with forward-looking activity signals



The Micro View should prioritize high CLTV × low probability alive, not historical Champions alone.